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Opposing effects of plant growth regulators via clonal integration on apical and basal performance in alligator weed

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Abstract

Invasive plants are a major threat to biodiversity and may adversely affect food security. Clonal integration enables the sharing of resources between connected ramets and can enhance plant performance in many invasive species. However, few studies have examined the role of clonal integration when weeds are exposed to plant growth regulators (PGRs). PGRs are used extensively in agriculture and may affect nearby weeds through soil leaching, erosion and runoff. Our aim was to investigate the effects of clonal integration on growth in a noxious weed, *Alternanthera philoxeroides* (alligator weed), in response to two PGRs frequently used in agriculture, gibberellins (GAs) and paclobutrazol (PAC). Ramets of *A. philoxeroides* were propagated in the greenhouse, and treated with PGRs. PGRs were applied to the older ramets (i.e. 'basal' part), with half of the plants having the stems between the apical (younger) and basal parts left connected, while the remaining plants had the stems between the two parts severed. Following the growing period, plants were measured for growth traits. We found that GA and PAC had contrasting effects on plant growth. GA significantly promoted above-ground growth of the apical ramets via clonal integration. Alternatively, PAC inhibited above-ground growth in the basal and apical parts, and enhanced below-ground growth of the basal and apical ramets through clonal integration. Our results highlight how clonal integration can promote growth in *A. philoxeroides* following the application of PGRs, which is likely an important mechanism for this species to invade new environments.

Keywords agricultural weed, exotic clonal plant, morphological traits, plant hormones, plant invasions

植物生长调节剂通过克隆整合对空心莲子草顶端和基部生长的不同作用

摘要：入侵植物不仅对全球生物多样性造成了巨大的威胁，同时也严重影响了农业生产与粮食安全。克隆整合使得相连植株进行资源共享，能促进入侵植物的生长从而获得优势。然而，入侵杂草在植物调节剂(plant growth regulators, PGRs)影响下的克隆整合作用则很少有报道。PGRs被广泛应用于

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农作物生产上，并能通过土壤淋溶、侵蚀和径流作用，影响分布在作物附近的农田杂草的生长。本研究采用两种PGRs赤霉素(gibberellins, GA)和多效唑(paclobutrazol，PAC)处理恶性入侵杂草空心莲子草(*Alternanthera philoxeroides*)基端，并保持或者通过剪切达到控制基端与顶端的连通，从而探究克隆整合作用在空心莲子草响应两种农业常用PGRs中的作用。研究结果表明，GA和PAC对空心莲子草生长的作用相反。GA通过克隆整合作用显著促进顶端植株的地上生长。相反地，PAC显著抑制基端和顶端的地上生长，但是能够通过克隆整合作用显著促进基端和顶端的地下生长。这些研究结果解释了克隆整合作用能促进PGRs对空心莲子草生长的促进作用，这很可能是外来杂草能够成功入侵人为干扰较多的农业生态系统的重要原因之一。

关键词：农田杂草，外来克隆植物，表型特征，植物激素，植物入侵

INTRODUCTION

Invasive plant species are a significant threat to global agriculture and can cause huge crop losses and adversely affect food security (Cook *et al.* 2011; Dai *et al.* 2020; Paini *et al.* 2016; Pimentel *et al.* 2001). Like invasive insects and pathogens, which are known to cause significant damage to crops and forest production (Pimentel *et al.* 2005), invasive weeds are considered a major threat to agriculture worldwide (Paini *et al.* 2016). Invasive plants possess many traits that have negative impacts on co-occurring species, including aggressive growth and strong competition for resources (e.g. water, nutrients and sunlight), and they often act as host plants for pests and diseases (Torres *et al.* 2013).

Invasive weeds are often clonal and clonal growth can enhance the invasion potential of many introduced weeds (Roiloa *et al.* 2019). A unique trait of clonal plants is clonal integration, which enables the sharing of resources and/or signals between connected ramets within the one clone (Yu *et al.* 2010). Numerous studies have shown that clonal integration can increase growth, resource acquisition and photosynthetic capacity of invasive clonal weeds (Dong *et al.* 2015; Liu *et al.* 2008; Wan *et al.* 2017; Wang *et al.* 2008; Xu *et al.* 2010; Yu *et al.* 2009). For example, the sharing of resources (e.g. water, nutrients and carbohydrates) via clonal integration has been found to increase biomass and plant performance of weeds in heterogeneous and variable environments, thereby conferring invasive plant species with a competitive advantage over natives (Song *et al.* 2013; Wang *et al.* 2017b). As such, clonal integration is widely considered to be an important mechanism in plant invasions (Chen *et al.* 2019; Song *et al.* 2013). However, although clonal integration of invasive plants across diverse and resource variable habitats has been well studied, few studies have explored the impacts of clonal integration when weeds are exposed to many of the chemicals utilized in agriculture (e.g. plant growth regulators [PGRs]).

PGRs are widely used in agroecosystems to increase crop yields and are important for boosting agricultural production (Rademacher 2015). Plant growth and development are affected by the addition of plant hormones such as gibberellins (GAs), cytokinins and auxins (Quilty and Cattle 2011). GAs play an important role in plant development, including seed germination, vegetative growth and reproduction (Swain and Singh 2005). GAs can promote lateral organ formation and differentiation and are commonly used in agriculture (Kong *et al.* 2016). In contrast, the PGR, paclobutrazol (PAC), inhibits the oxidation of *ent*-kaurene to *ent*-kaurenoic acid thereby suppressing GA production (Hedden and Graebe 1985). PAC reduces growth in many plant species (Cohen *et al.* 2019) and has been successfully applied in rice fields, where it is used to suppress plant height, increases plant compactness and prevents lodging (Jiang and Asami 2018; Sharma and Awasthi 2005). When applied to crop plants, PGRs may be either sprayed directly onto weeds that have encroached onto farmland, or seep via soil erosion or rainfall (Wu *et al.* 2013; Zhao *et al.* 2007) and are then absorbed by nearby invasive plants (the latter scenario is more likely in most agricultural ecosystems). PGRs could therefore greatly impact the growth and competition of weeds in or near crop fields, which may pose a serious threat to agricultural production. Previous research has revealed that chemicals applied to agricultural ecosystems can promote plant invasions. For example, over the last few decades, fertilization has been reported to impact weed abundance, density and diversity (Moss *et al.* 2004; Wan *et al.* 2012). In fact, fertilizer has been found to directly promote

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growth and biomass in clonal weeds through clonal integration (Huang *et al.* 2018). Yet, the effects of PGRs on plant performance (i.e. growth and survival, Poorter and Bongers 2006; Poorter *et al.* 2018) via clonal integration is unknown for the majority of invasive plant species.

Alternanthera philoxeroides (commonly known as alligator weed) is an amphibious, perennial clonal herb native to South America (Souza *et al.* 2015). The species has become a notorious invasive weed globally, spreading to over 30 countries (Geng *et al.* 2006), and has caused large-scale economic and ecological problems (Ding *et al.* 2007; Pandya *et al.* 2015; Zhang *et al.* 2020). In China, *A. philoxeroides* spreads mainly vegetatively and rarely produces viable seeds. Clonal integration has been found to enhance growth in *A. philoxeroides* in both homogeneous and heterogeneous environments (Dong *et al.* 2015; Liu *et al.* 2008). *Alternanthera philoxeroides* may also benefit from clonal integration when it is exposed to disturbance or injury (e.g. defoliation, You *et al.* 2014b). As such, clonal integration is thought to have facilitated the expansion of *A. philoxeroides* into new environments (Wang *et al.* 2009).

Alternanthera philoxeroides has invaded many ecosystems, including croplands, lawns, gardens, marshes and lakes (Deng *et al.* 2006; Xu *et al.* 2009). The species is particularly aggressive in paddy farms (Fig. 1) and is recognized as a noxious weed in rice fields in 23 provinces of China (Mahajan and Timsina 2011). Populations of *A. philoxeroides* in rice fields are frequently exposed to PGRs, including GA and PAC, during the growing season. Yet, the impacts of such PGRs on *A. philoxeroides* have been understudied. In a previous study, the application of the PGRs, aviglycine and naphthalene acetic acid, increased below-ground biomass and the number of younger growing tips in *A. philoxeroides* (Clements *et al.* 2012). However, given that GA and PAC can have different impacts on plant growth, we explored the effects of these two PGRs on *A. philoxeroides* in this study. Additionally, since clonal integration has been demonstrated to play an important role in the performance and spread of *A. philoxeroides*, it is likely that the impacts of PGRs on plant growth in this species will be influenced by resource sharing between ramets. Thus, we also examined the effects of PGRs on growth of different ramets (i.e. older and younger nodes) of *A. philoxeroides* and how this is affected by clonal integration.

The aim of the current study was to examine the impacts of clonal integration on growth traits of *A. philoxeroide*s in response to the application of GA and PAC (two important PGRs commonly used in rice fields). Specifically, we tested the following hypotheses: (i) growth in *A. philoxeroides* will respond differently to these two PGRs and

Figure 1: *Alternanthera philoxeroides* in a paddy field in Fujian province, China (photograph by Zhicong Dai).

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(ii) growth in connected ramets will be affected by the application of these PGRs through clonal integration.

MATERIALS AND METHODS

Plant collection, propagation and treatments

Populations of invasive *A. philoxeroides* in China have been found to be very low in genetic diversity (Wang *et al.* 2005). However, the species may display high phenotypic plasticity across environments. To avoid different phenotypes affecting the experiment results in our study, clonal fragments from one population of *A. philoxeroides* were collected from a wetland in Zhenjiang, Jiangsu Province, China (119°31.76′ E, 32°12.02′ N), and vegetatively propagated in a greenhouse at Jiangsu University. Healthy and similar-sized clonal fragments (with four nodes and a stolon apex) were surface sterilized in 5% NaClO solution for 10 min, and then washed five times with sterile deionized water. The two older nodes (hereafter referred to as the 'basal part') of the fragments were planted in 60 plastic pots (10 cm × 10 cm × 8 cm) containing 100 g of sand and vermiculite (weight ratio: 2:1). The two younger nodes (hereafter referred to as the 'apical part') of the fragments were planted into another 60 plastic pots with the same dimensions and soil mixture.

To investigate the effects of PGRs on the growth of *A. philoxeroides*, 30 μmol L^{-1} of GAs and 30 μmol L^{-1} PAC were applied separately to the basal parts of 20 plants on the third day after planting. These concentrations of GA and PAC were used as they are utilized in rice fields to increase crop yields (Yang *et al.* 2012) and prevent lodging (Du *et al.* 2020), respectively. We also performed preliminary experiments to confirm that these concentrations of the PGRs affected plant growth in *A. philoxeroides*. Both PGRs were purchased from a commercial company (Sangon Biotech, Shanghai, China). Before applying the PGRs in our experiment, we considered the exposure of *A. philoxeroides* to GA and PAC in rice fields under natural conditions. When PGRs are applied to rice plantations, they often do not directly come into contact with weeds such as *A. philoxeroides* because they are usually applied in the early stages of rice planting. However, weeds may be exposed to PGRs through rainfall, soil seepage and erosion. As such, the basal parts of the plants (i.e. the older parts rooted in the soil) are more likely to come into contact with PGRs than the younger (apical) parts. Therefore, for our study we applied PGRs to the basal parts of the clonal fragments to simulate what likely happens under natural conditions. For the control treatment, purified water was added to the basal parts of the remaining 20 plants.

To test the role of clonal integration, the stems between the basal and apical parts of half of the plants in each treatment were left connected. The remaining plants had the stems between the basal and apical parts carefully severed. All severed clonal fragments were disinfected with 5% NaClO to prevent infection. There were six treatment combinations in our experiment: three plant hormone regulator treatments (i.e. GA, PAC and control) × two status of clonal integration (i.e. connected or disconnected). There were 10 replications in each treatment combination. All clonal fragments were randomly arranged in the greenhouse and rotated once per week. Full-strength Hoagland solution was added to the clonal fragments every week to meet plant nutritional needs. Purified water was added to all the plants when needed.

Growth trait measurements

After 2 months of growth, all plants were harvested and growth characters were measured for apical and basal parts of each clonal fragment, including the main shoot length, above-ground dry mass, node number of the main shoot, leaf area, root length and below-ground dry mass. Shoot length was measured to the nearest mm from the axil of the main shoot to the shoot tip using a ruler. Leaf area was determined for two fully expanded leaves per plant (i.e. the third pair of leaves from the tip of the main shoot). Each leaf pair was scanned and the leaf area (to the nearest mm^2) was calculated using ImageJ (Qi *et al.* 2020). Roots from each clonal fragment were carefully removed from the soil using a 2 mm sieve, and root length (to the nearest mm) was measured using a ruler. Both above-ground and below-ground plant material for each clonal fragment were dried in an oven to constant mass at 80 °C for 72 h and dry mass determined. We calculated root-to-shoot ratio (root:shoot ratio) for the apical and basal parts (determined as below-ground dry mass to above-ground dry mass), apical total dry mass (i.e. above- and below-ground dry mass of the apical part), basal total dry mass (above- and below-ground dry mass of the basal part) and total biomass of the whole clone (apical + basal parts).

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Data analysis

We used a two-way analysis of variance (ANOVA) to assess the main effects of PGRs and clonal integration, and interactions between these effects on plant growth traits in *A. philoxeroides*. Tukey's honest significance test (HSD) was performed to investigate differences in all growth traits among the PGRs and clonal integration treatments. Analyses were performed separately for basal and apical clonal fragments, as well as on total biomass of the whole clone (of apical and basal ramets combined). To examine whether the data met the assumptions of ANOVA, we tested all data for normality using the Shapiro–Wilks test and for equality of variance with Levene's test prior to each analysis. Traits were $\log_{10}$ transformed where necessary to meet the assumptions of ANOVA. For some traits, the $\log_{10}$ transformed data did not meet the assumptions of ANOVA, and in such cases, we used the Scheirer–Ray–Hare test, which is an extension of the Kruskal–Wallis test and is a nonparametric equivalent to a two-way ANOVA (Mukherjee and Blaustein 2019; Rooke and Bergstrom 2007; Scheirer *et al.* 1976; Sokal and Rohlf 1995). Dunn's nonparametric multiple comparison test was then used for all *post hoc* evaluations (Sokal and Rohlf 1995). All analyses were performed in R version 3.6.1. Data were tested at the significance level $P < 0.05$ in all our analyses.

RESULTS

Our findings revealed that most growth traits in both the basal and apical parts of *A. philoxeroides* were significantly affected by the PGRs ($P < 0.05$, Table 1). Only leaf area, total dry mass and root length in the apical part, and below-ground dry mass in the basal part were not significantly affected by the PGRs ($P > 0.05$, Table 1). Clonal integration significantly affected above- and below-ground dry mass, total dry mass and root:shoot ratio in the basal part of the plants ($P < 0.05$, Table 1). In the apical part, only leaf area and above-ground dry mass were significantly affected by clonal integration ($P < 0.05$, Table 1). Apart from below-ground dry mass and root length, all other traits in the apical part were significantly affected by the interaction between clonal integration and PGRs ($P < 0.05$, Table 1). In contrast, below-ground dry mass and root length in the basal part were significantly affected by the interaction between clonal integration and PGRs ($P < 0.05$, Table 1). Therefore, the interaction between PGRs and clonal integration significantly affected above-ground growth in the apical parts and below-ground growth in the basal parts. The total biomass of the whole clone (i.e. apical and basal parts) was significantly affected by PGRs, clonal integration and the interaction between PGRs and clonal integration ($P < 0.05$, Table 1).

Effects of PGRs and clonal integration on apical and basal dry mass and total biomass of the whole clone

We found that the application of GA promoted the total dry mass of connected apical ramets through clonal integration (Fig. 2a). The total dry mass of connected apical ramets in the GA treatment was significantly greater than the total dry mass of disconnected apical ramets in the same treatment ($P < 0.05$, Fig. 2a). Alternatively, the application of PAC significantly inhibited the total dry mass of the disconnected basal ramets (compared with all other treatments, $P < 0.05$), but did not significantly affect the basal connected ramets compared with ramets in the other treatments (Fig. 2a). For the whole clone (i.e. apical and basal parts combined), we found the application of PAC inhibited total biomass of disconnected ramets (Fig. 2b). The total biomass of the disconnected ramets in the PAC treatment was significantly lower compared with the disconnected ramets in the control and connected ramets in the GA treatments, respectively ($P < 0.05$, Fig. 2b). In contrast, the total biomass of connected ramets in the PAC treatment was not significantly different from connected or disconnected ramets in the other treatments ($P > 0.05$, Fig. 2b).

Effects of PGRs and clonal integration on above-ground performance

In the apical part, shoot length, node number, leaf area and above-ground dry mass of the connected stems were significantly greater than in plants with disconnected stems when GA was applied ($P < 0.05$, Fig. 3). Additionally, node number of connected apical ramets in the GA treatment was significantly higher compared with both connected and disconnected apical ramets in the control and PAC treatments ($P < 0.05$, Fig. 3b). In contrast, most above-ground growth traits in the basal part were not significantly affected by the application of GA (Fig. 3). Only above-ground dry mass of the disconnected basal ramets was significantly higher than the connected basal ramets in the GA treatment ($P < 0.05$, Fig. 3d).

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Table 1: The effects of PGRs and clonal integration and interactions between these effects on the growth of *Alternanthera philoxeroides* ($n = 60$)

(a) ANOVA										
		PGR			Clonal integration (CI)			PGR × CI		
Part	**Trait**	*d.f.*	*F*	*P*	*d.f.*	*F*	*P*	*d.f.*	*F*	*P*
Apical	Leaf area	2	1.350	0.278	1	13.470	**0.001**	2	4.930	**0.016**
	Above-ground dry mass	2	5.917	**0.005**	1	3.654	**0.061**	2	8.082	**<0.001**
	Below-ground dry mass	2	3.489	**0.038**	1	0.009	0.924	2	1.910	0.158
	Total apical dry mass	2	0.951	0.393	1	2.709	0.106	2	5.489	**<0.001**
	Root-to-shoot	2	7.834	**0.001**	1	1.950	0.168	2	3.758	**0.03**
Basal	Leaf area	2	16.665	**<0.001**	1	0.916	0.348	2	0.257	0.775
	Root length	2	4.500	**0.016**	1	0.445	0.507	2	6.903	**0.002**
	Below-ground dry mass	2	0.717	0.493	1	41.617	**<0.001**	2	14.135	**<0.001**
	Total basal dry mass	2	11.593	**<0.001**	1	8.355	**0.006**	2	14.545	**<0.001**
	Root-to-shoot	2	16.270	**<0.001**	1	63.960	**<0.001**	2	3.700	**0.032**
(b) Scheirer–Ray–Hare										
		PGR			Clonal integration (CI)			PGR × CI		
Part	**Trait**	*d.f.*	*H*	*P*	*d.f.*	*H*	*P*	*d.f.*	*H*	*P*
Apical	Shoot length	2	8.517	**0.014**	1	1.544	0.214	2	21.493	**<0.001**
	Node number	2	9.703	**0.008**	1	0.802	0.371	2	16.721	**<0.001**
	Root length	2	3.751	0.153	1	0.542	0.462	2	2.615	0.271
Basal	Shoot length	2	34.678	**<0.001**	1	1.524	0.217	2	1.755	0.416
	Node number	2	30.393	**<0.001**	1	0.810	0.368	2	1.010	0.603
	Above-ground dry mass	2	22.830	**<0.001**	1	8.282	**0.004**	2	0.955	0.62
Whole clone	Total biomass (apical and basal)	2	9.336	**0.009**	1	4.466	**0.035**	2	6.530	**0.038**

Effects on plant traits were determined using: (**a**) two-way ANOVA and (**b**) Scheirer–Ray–Hare tests. Abbreviations: *d.f.* = degrees of freedom, *P* = significance. Significant *P* values are in bold.

Conversely, PAC application suppressed above-ground growth of the basal ramets. Shoot length and node number of connected and disconnected basal ramets in the PAC treatment were significantly lower compared with all basal ramets in the GA and control treatments ($P < 0.05$, Fig. 3a and c). Similarly, above-ground dry mass of the connected basal ramets was significantly lower after being treated with PAC compared with connected and disconnected basal ramets in the control treatment ($P < 0.05$, Fig. 3d). Shoot length and node number of connected ramets in the apical parts were also significantly inhibited by PAC compared with connected apical ramets in the GA and control treatments ($P < 0.05$, Fig. 3a and c). Moreover, shoot length and node number of connected apical stems were significantly lower than in disconnected apical stems when PAC was applied ($P < 0.05$, Fig. 3a and c).

Effects of PGRs and clonal integration on below-ground performance and root-to-shoot ratio

Our findings revealed that the application of GA resulted in significantly longer roots in disconnected basal ramets compared with connected basal ramets

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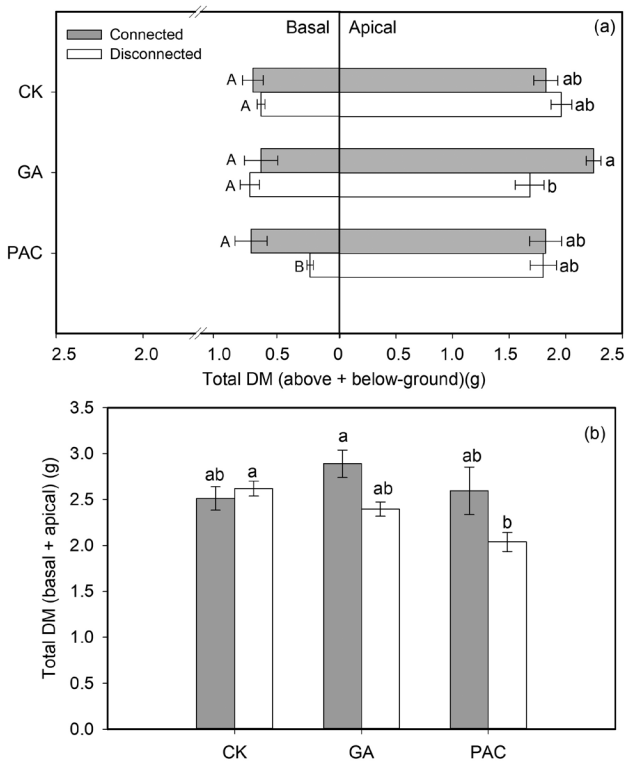

Figure 2: Effect of PGRs (GA and PAC) and severance on growth in *Alternanthera philoxeroides* showing (**a**) total dry mass of basal (left bars) and apical (right bars) parts and (**b**) total biomass (of both basal and apical parts). CK represents the control treatment (where only purified water was added). Bars show mean ± SE ($n = 10$). Different letters indicate significant differences ($P < 0.05$) among the six treatments of PGRs and severance or connection of internodes.

in the control ($P < 0.05$, Fig. 4a). In contrast, GA treatment had no significant effect on basal below-ground dry mass in connected or disconnected stems compared with the control (Fig. 4b). Moreover, below-ground traits in the apical part were not affected by the application of GA, with neither root length nor below-ground dry mass of connected or disconnected apical stems being significantly different between the GA and control treatments (Fig. 4a and b).

Alternatively, below-ground growth traits of the basal part of plants were significantly affected by the application of PAC. In the PAC treatment, roots of connected basal ramets were significantly longer than connected stems in the control treatment ($P < 0.05$, Fig. 4a). Furthermore, in the PAC treatment, root length of the connected basal stems was significantly greater compared with disconnected ramets ($P < 0.05$, Fig. 4a). Below-ground dry mass of disconnected basal ramets was significantly lower after being treated with PAC compared with all basal ramets in the other treatments ($P < 0.05$, Fig. 4b). Additionally, in the PAC treatment, the disconnected basal ramets had significantly lower below-ground dry mass than the connected basal ramets ($P < 0.05$, Fig. 4b). In the apical part, only below-ground dry mass of connected ramets in the PAC treatment was significantly greater than connected ramets in the control treatment (Fig. 4b).

PAC significantly increased root:shoot ratio in the basal ramets through clonal integration (Fig. 4c). Root:shoot ratio of the connected basal ramets in the PAC treatment was significantly greater than the disconnected basal ramets in the same treatment, and all ramets in the other treatments ($P < 0.05$, Fig. 4c). In the apical part, root:shoot ratio of connected ramets in the PAC treatment was significantly greater than that of the connected apical ramets in the GA and control treatments ($P < 0.05$), but was not significantly different from the disconnected apical ramets in any of the three treatments (Fig. 4c).

DISCUSSION

Consistent with our first hypothesis, PGRs significantly affected growth in the invasive weed, *A. philoxeroides* (Table 1). Each PGR affected different parts of the clonal weed, with GA promoting apical above-ground growth, PAC inhibiting basal above-ground growth, and PAC increasing apical and basal below-ground growth (Figs 3 and 4). Growth was promoted in ramets that were not directly in contact with PGRs and this growth appeared to be facilitated by clonal integration. Our findings revealed for the first time that GA and PAC have different effects in different ramet parts, and clonal integration could transfer the resources produced from the application of these PGRs from basal to apical parts (see Fig. 5 for a model showing the role of PGRs and resource sharing via clonal integration in this species).

The effects of GA on growth in *A. philoxeroides* via clonal integration

PGRs are an important component in agricultural production and play a major role in modifying crop development, stress response and crop yields, and have been used all over the world on a diverse range of crops (Abdelgadir *et al.* 2009; Anjum *et al.* 2011). GA is important for promoting plant growth (particularly shoot elongation; Hedden 2016; Swain and Singh 2005) and this was supported by the findings of our study, which revealed that this

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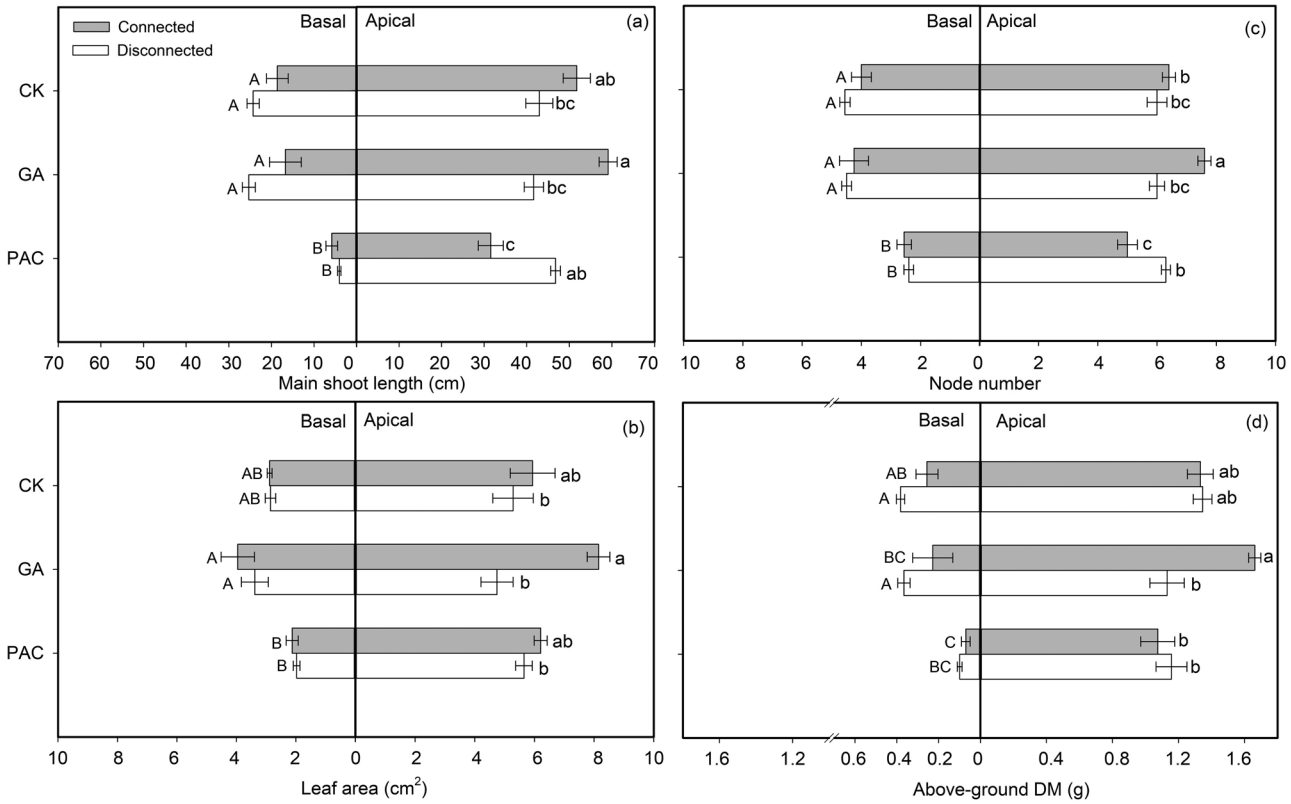

Figure 3: Effect of PGRs (GA and PAC) and severance on (**a**) shoot length, (**b**) leaf area, (**c**) node number and (**d**) above-ground dry mass (DM) of basal parts (left bars) and apical parts (right bars) of ramets of *Alternanthera philoxeroides*. CK represents the control treatment. Bars show mean ± SE (n = 10). Different letters indicate significant difference ($P < 0.05$) among the six treatments of PGRs and severance or connection of internodes.

hormone significantly enhanced the majority of above-ground growth traits in connected stems in the apical parts of *A. philoxeroides* (compared with disconnected ramets in the control, Fig. 3). However, we also found that GA had no significant effect on most above-ground traits in the basal stems compared with the control (Fig. 3). In fact, our results revealed that only root growth of the disconnected basal ramets in the GA treatment was significantly higher than in the connected basal ramets in the control (Fig. 4a). This finding agrees with the results of Singh *et al.* (2019) who showed that GA promoted root growth in sweet potato. Thus, the effects of GA on above- and below-ground growth may be species specific, and additional research on more species would likely provide further insights into the role and mechanism of GA movement in plant growth (Gupta and Chakrabarty 2013).

As predicted by our second hypothesis, we found that growth in *A. philoxeroides* was affected by GA through clonal integration. For example, all above-ground traits of the connected apical ramets were enhanced by the application of GA (compared with the disconnected apical ramets in the same treatment, Fig. 3). Previous studies have shown that plants may allocate nutrients to younger parts via clonal integration to maintain their function (Adomako *et al.* 2020; He *et al.* 2011). This research agrees with our study, where resources appeared to have been transferred from the basal part of the plant to the apical part of the plant. However, from our findings it was not possible to ascertain whether the plant hormone GA was directly transferred from the basal ramets to the apical ramets, or if GA increased the plant's ability to uptake water and nutrients from the soil (which was then transferred to the apical part of the plant). A study examining the plant hormone content of apical and basal ramets following the application of the PGRs would offer further insights into which of these two mechanisms are occurring in *A. philoxeroides*.

The effects of PAC on growth in *A. philoxeroides* via clonal integration

Our findings revealed that unlike GA, PAC significantly inhibited most above-ground basal

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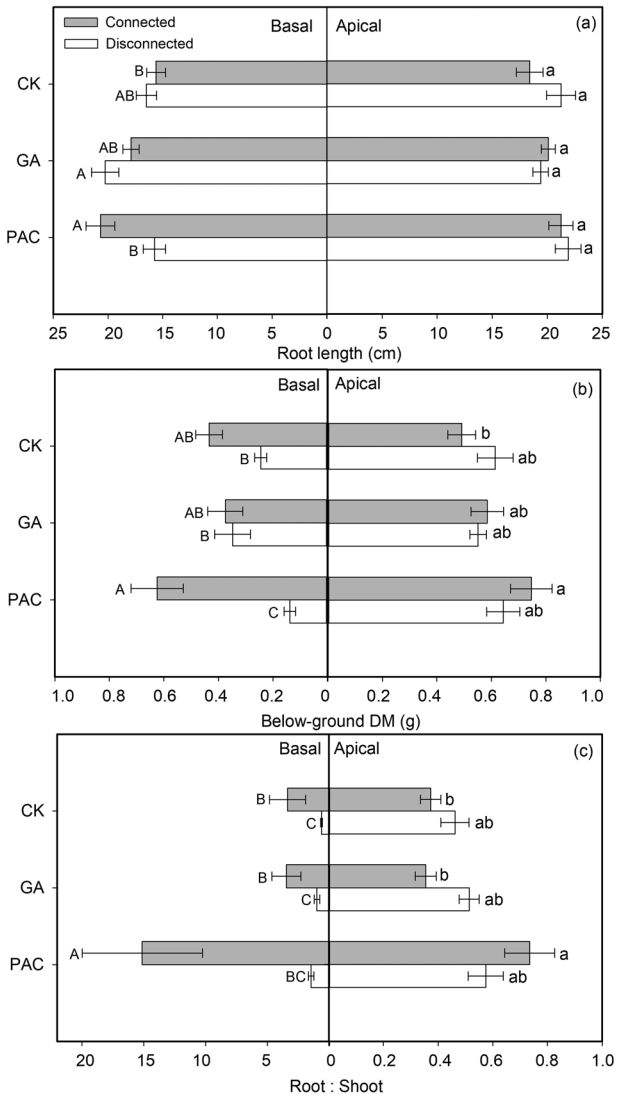

Figure 4: Effects of PGRs (GA and PAC) and severance on (**a**) root length, (**b**) below-ground dry mass (DM) of basal parts (left bars) and apical parts (right bars) and (**c**) root-to-shoot (root:shoot) of basal (left bars) and apical (right bars) parts of ramets of *Alternanthera philoxeroides*. CK represents the control treatment. Bars show mean ± SE ($n = 10$). Different letters indicate significant differences ($P < 0.05$) among the six treatments of PGRs and severance or connection of internodes.

growth traits in *A. philoxeroides*, and had mostly a nonsignificant effect on above-ground apical growth (Fig. 3). However, we also found that PAC significantly promoted below-ground basal growth through clonal integration (Fig. 4). The finding of PAC promoting basal below-ground growth is in contrast to a previous study where both above- and below-ground growth in the invasive weed, *Sphagneticola trilobata*, was inhibited by PAC (Dai *et al.* 2016). However, the effects of PAC on below-ground growth have been found to differ across species (Kamran *et al.* 2018; Liu *et al.* 2021). For example, Liu *et al.* (2021) found that root mass was reduced by PAC in invasive plants, but was increased in native plants. Therefore, as with GA, the role of PAC in plant growth may also be species specific. When plants are under stress, they allocate a greater proportion of their biomass to the root system (Hermans *et al.* 2006). In the case of *A. philoxeroides*, although PAC inhibited above-ground growth, resources might be allocated by the plant to the roots to compensate for this inhibition in above-ground growth, thereby resulting in a higher root:shoot ratio (Fig. 4c). Consequently, our findings suggest that this compensation response together with clonal integration, contributed to enhanced below-ground growth of the basal ramets after being treated with PAC.

The results from the present study suggest that the application of PAC has a stronger effect on growth in *A. philoxeroides* than GA. For instance, when compared with the control treatment, PAC significantly inhibited many above-ground growth traits (i.e. shoot length, node number and above-ground dry mass) of connected and disconnected basal ramets (Fig. 3). In contrast, compared with the control, GA only significantly increased node number of apical connected ramets (Fig. 3). A number of biochemical effects have been documented in plants treated with PAC, such as decreases in GA levels (Bultynck and Lambers 2010; Rademacher 2000) and reduced ethylene production (which is associated with promoting vegetative growth and regulating senescence, Iqbal *et al.* 2017; Wang and Steffens 1985). Therefore, the inhibition of PAC on plant growth in our study is not surprising since the application of this PGR probably affected such biochemical processes in *A. philoxeroides*. Moreover, although it is likely that *A. philoxeroides* is highly sensitive to a reduction in endogenous GA (induced by the application of PAC), the species may be less responsive when exogenous GA is applied. Levels of endogenous GA are likely to be particularly important for *A. philoxeroides* when the species encounters biotic or abiotic stresses. However, more research on the effects of endogenous and exogenous GA on growth in *A. philoxeroides* are required before conclusions can be drawn.

PGRs and clonal integration in plant invasion

Previous studies suggest that invasive plants benefit more from GA than native plants (Liu *et al.* 2021). In

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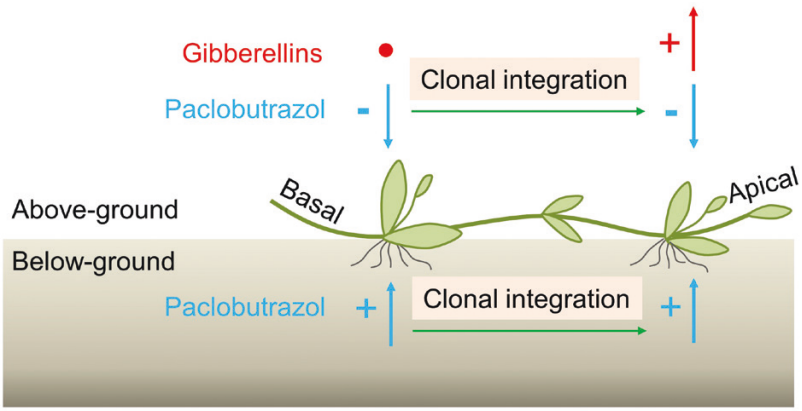

Figure 5: A model of the roles of PGRs and clonal integration on growth of the agricultural clonal weed, *Alternanthera philoxeroides*. Different PGRs have contrasting effects on growth in different parts of *A. philoxeroides*. GAs have no significant effects on above-ground growth in the basal part of the plant, but promotes above-ground growth of the apical part indirectly through clonal integration compared with the disconnected stems. PAC directly inhibits above-ground growth in the basal part, and also inhibits above-ground growth in the apical part. However, PAC promotes below-ground growth in the basal part, and also indirectly promotes below-ground growth of the apical part through clonal integration.

the current study, we found that both GA and PAC promoted growth in different parts of the invasive weed, *A. philoxeroides*, and this growth was often facilitated by clonal integration. Clonal integration has been found to facilitate invasion success of many exotic clonal species following introduction into new habitats (Pu *et al.* 2022; Wang *et al.* 2009). For example, the results of Wang *et al.* (2017b) revealed that clonal integration confers a competitive advantage to invasive plants (by producing more biomass) over natives. In *A. philoxeroides*, clonal integration has been found to increase plant performance at different water depths (Dong *et al.* 2015), and may alter the allocation of resources under varying nitrogen availability (Wang *et al.* 2017a). Thus, increased growth facilitated by clonal integration is likely to be an important mechanism for invasive plants, such as *A. philoxeroides*, in competitive, resource-patchy and/or stressful environments (e.g. You *et al.* 2014a).

Our findings suggest that the promotion of growth in *A. philoxeroides* via the application of PGRs and clonal integration together could increase the competitive ability of this species. However, to truly appreciate the impacts of PGRs and clonal integration on this species and how it impacts interspecific competition would require a study that also examines the effects of PGRs on native species that commonly co-occur with invasive *A. philoxeroides*. Another limitation of our study was that in order to examine the effects of clonal integration, it was necessary for us to sever stems for the disconnected treatment. While we took care that ramets were severed carefully, and the cut stems were disinfected, a wound was introduced to the plant through this process. As such, it is possible that our results have overestimated the role of clonal integration (an issue rarely acknowledged in the literature). However, since *A. philoxeroides* is a clonal plant, it was necessary for us to cut the ramets for all the treatments (and so all stems in every treatment had cut stems). Thus, the findings across treatments in our study should be comparable. Nevertheless, additional research examining the effects of severing stems on clonal integration and the sharing of resources between ramets would provide further insights into this issue.

Although PGRs are more likely to be exposed to the basal ramets of *A. philoxeroides* under natural conditions, the effects on growth of this species when PGRs are directly applied to the apical parts is currently unknown. Given that under certain conditions the apical part of *A. philoxeroides* may be exposed directly to PGRs (e.g. if the weed encroaches onto farmland at the early stage of rice planting), this is an interesting area of research that could be explored in the future. Furthermore, the impacts of the application of PGRs and clonal integration in invasive species on nearby crop species provides a promising avenue for future research, and is likely to be informative for planning for the prevention and control of invasive clonal species in agricultural ecosystems.

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CONCLUSIONS

The impacts of clonal integration on the growth of invasive plants when they are exposed to different PGRs have rarely been reported. In this study, we found that GA and PAC have contrasting effects on the growth of a noxious global weed (*A. philoxeroides*); with the former promoting above-ground growth of apical ramets, and the latter inhibiting many basal above-ground growth traits but enhancing below-ground growth. Thus, it would appear that the application of these two PGRs induced a division of tasks in the apical and basal ramets of the plant. The effects of the PGRs on growth in *A. philoxeroides* were also affected by clonal integration. Therefore, our findings suggest that the impacts of PGRs together with clonal integration enhance growth in *A. philoxeroides*, and could increase the expansion ability of this species in areas where PGRs are often used.

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Conflict of interest statement. The authors declare that they have no conflict of interest.

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